

Programming Documentation

CommandBrick - Capacitive Bluetooth Keyboard and Mouse

1. Overview

The CommandBrick software was developed for the ESP32-S3 using the Arduino Framework. The software combines multiple hardware components into a single Bluetooth Human Interface Device (HID), allowing the user to control a computer using both keyboard and mouse functionality.

The ESP32 continuously communicates with the connected sensors over the I²C bus, processes the incoming data, and transmits the corresponding Bluetooth HID reports to the connected computer.

The software consists of four major modules:

- Hardware initialization
- Bluetooth HID communication
- Mouse control
- Keyboard control

2. Development Environment

IDE: Arduino IDE

Board: ESP32-S3

Framework: Arduino Framework

Communication: I²C

Bluetooth: BLE HID

3. Used Libraries

- **Wire.h** - I²C communication
- **NimBLEDevice.h** - Bluetooth Low Energy stack
- **NimBLEServer.h** - BLE server implementation
- **NimBLEHIDDevice.h** - Bluetooth HID keyboard and mouse functionality

4. Software Architecture

The main loop repeatedly performs the following tasks:

1. Read touchpad data
2. Read capacitive button states
3. Process user input
4. Send Bluetooth reports
5. Generate vibration feedback if necessary

5. Hardware Initialization

During startup the software configures the GPIO pins, initializes the I²C bus, configures Bluetooth HID, initializes the MPR121 capacitive touch controller, initializes the Cirque Pinnacle touchpad, configures the vibration motor, and starts BLE advertising.

6. Bluetooth HID Implementation

The ESP32-S3 operates as a Bluetooth Human Interface Device implementing two HID reports:

- Keyboard Report
- Mouse Report

This allows the device to be recognized by the operating system without additional drivers.

7. Mouse Control

Mouse movement is generated using the Cirque Pinnacle touchpad. The software reads X/Y movement over I²C, applies dead-zone filtering and cursor acceleration, and sends Bluetooth mouse reports.

8. Keyboard Control

Keyboard input is generated using the MPR121 capacitive touch controller.

Button Mapping:

- 0 - Left Mouse Click
- 1 - Right Mouse Click
- 2 - Space
- 3 - Enter
- 4 - Tab
- 5 - Backspace
- 6 - Ctrl
- 7 - Alt
- 8 - Shift
- 9 - Arrow Up
- 10 - Arrow Down
- 11 - Open On-Screen Keyboard (Ctrl + GUI + O)

9. Vibration Feedback

Whenever a capacitive button is activated, the ESP32 briefly enables the vibration motor connected to GPIO 5, providing tactile confirmation to the user.

10. Main Program Flow

Initialize Hardware → Initialize Bluetooth → Initialize MPR121 → Initialize Touchpad → Start BLE Advertising → Wait for Connection → Read Touchpad → Read Capacitive Buttons → Process Input → Send HID Reports → Repeat

11. Communication

The sensors share the same I²C bus.

MPR121 Address: 0x5A

Cirque Pinnacle Address TM035035: 0x2A

12. Program Features

- Bluetooth mouse control
- Bluetooth keyboard control
- Capacitive touch sensing
- Touchpad cursor movement
- Vibration feedback
- Automatic Bluetooth pairing
- Low-power BLE communication

13. Possible Future Improvements

- Configurable button mappings
- Adjustable cursor sensitivity
- Gesture recognition
- Battery level monitoring
- OTA firmware updates
- Deep sleep mode